

Roll Number: _____	
Thapar Institute of Engineering & Technology, Patiala	
Department of Energy & Environment	
B.E. 1st Yr EST	UEN008/UEN002: Energy & Environment
20 December, 2024	Name of Faculty: Amit Dhir, Anoop Verma, Gaurav Goel, Shilpi Verma, Ashutosh Kumar, Nishu Joshi
Time: 03 Hours; MM: 80	
Note: Attempt <u>Any five</u> questions. Assume missing data, if necessary.	

Q. No.	Questions	Marks	CO	BL														
Q1.	a) Explain the working principle of a trickling filter used in biological treatment. Describe the key design considerations and execution steps involved in constructing and operating a trickling filter. Support your answer with a neat and labelled diagram.	(5)	3	L3														
	b) Define the concept of a circular economy. Discuss how the circular economy affects industrial activity. Use examples to illustrate how industries are transitioning to circular economy models.	(3)	1	L2														
	c) The BOD ₅ of a wastewater is 120 mg/L and the BOD rate constant is 0.115 day ⁻¹ at 20°C. What is the ultimate BOD? If the wastewater is incubated at 15°C with supply of oxygen, how much oxygen will be used by microorganisms in three days? Consider Θ value as 1.135.	(8)	2	L3														
Q2.	a) Explain the process of municipal solid waste (MSW) incineration and its role in waste management. Describe the design and execution of an incineration plant, highlighting the key components involved. Discuss the role of Air Pollution Control Devices (APCDs) in ensuring environmentally compliant incineration. Support your answer with a diagram of an MSW incineration system.	(8)	2	L2														
	b) A residential waste has the following components. Estimate the moisture content of the mixed waste sample. <table><tr><th>Component</th><th>Percentage content</th><th>Moisture content</th></tr><tr><td>Paper</td><td>50%</td><td>6%</td></tr><tr><td>Glass</td><td>20%</td><td>2%</td></tr><tr><td>Food</td><td>20%</td><td>70%</td></tr><tr><td>Yard Waste</td><td>10%</td><td>60%</td></tr></table>	Component	Percentage content	Moisture content	Paper	50%	6%	Glass	20%	2%	Food	20%	70%	Yard Waste	10%	60%	(8)	2
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Q3.	a) How does the solar thermal route facilitate the conversion of solar energy into heat? Discuss the principle, design, and applications of various solar thermal systems, with a focus on flat plate collectors and concentrating collectors with appropriate schematic diagrams.	(8)	4	L2														

	b)	A home in Tucson, Arizona, needs 62 kWh of heat on a winter day to maintain an indoor temperature of 20°C using solar thermal system. The specific heat capacity of water is 1 kcal/kg/°C, and the temperature difference between the hot fluid coming out of the system and the cold water entering the storage tank is 40°C. The average solar radiation in Tucson during the winter is 6.5 kWh/m ² /day. (a) What collector surface area is required for an all-solar heating system with 20% efficiency? (b) What size should the storage tank be to store enough energy for this requirement?	(8)	4	L4
Q4.	a)	State differences in the operating conditions between pyrolysis and gasification and its effect on the products produced. Also compare environmental impacts of these processes over conventional combustion & incineration processes?	(8)	4	L2
	b)	A biomass gasification plant processes 100 tonne of paddy straw with a generation of syngas 100 m ³ /tonne. The paddy straw has a carbon content of 40% by weight, and for every kilogram of carbon, 3.67 kg of CO ₂ is produced during oxidation. The biomass undergoes gasification at 1000°C, to produce syngas. For the purpose of this calculation, neglect any heat losses and unreacted carbon. Calculate total syngas produced in m ³ . What are the percentage of CO ₂ emissions relative to the total biomass processed? Assuming that 80% of the syngas is used for power generation, calculate the total electrical energy output (in kWh) if the syngas has an energy content of 40 MJ/kg. [Given density of syngas 1.1 kg/m ³]	(8)	4	L4
Q5.	a)	With the help of neat sketch, discuss the working principle of Bag house filter. Discuss its advantages and limitations.	(8)	3	L2
	b)	Design a parallel plate type electrostatic precipitator with 10 channels to handle 10,000 m ³ /h of gas for efficiency of 99%. Migration velocity of particles towards plate may be taken as 0.1m/s. Assume height of each plate to be 2m and let spacing between plates to be 0.15m.	(8)	3	L4
Q6.	a)	What do you mean by stability of atmosphere? Classify different stability class of atmosphere with the help of relative ELR and PLR values and graphs.	(8)	2	L2
	b)	An Ore refining industry is using a liquid fuel mixture of 60% Xylene (C ₈ H ₁₀) and 40%Toluene (C ₇ H ₈) for combustion. Find the (i) stoichiometric air to fuel ratio, and (ii) calculate the mass of CO ₂ production during combustion of 500g Xylene?	(8)	2	L3
